HW · 7 Jobson John Thomas

1) Given:

(a). cold stream in tubes : $\dot{m}c_{p,c} = 5000$ BTU/h · °F

$T_{c,i} = 90°F$, $T_{c,o} = 210°F$

Hot stream in shell : $\dot{m}c_{p,h} = 10000$ BTU/h · °F

$T_{hi} = 260°F$, one shell pass, one tube pass

Tubes = $D_o = 0.5$ in $= 0.041667$ ft

$N_t = 30$ tubes (single pass)

$U_o = 150$ BTU/h · ft^2 · °F

$$Q = \dot{m}c_{p,c}(T_{c,o} - T_{c,i}) = 5000(210 - 90)$$
$$= 6.00 \times 10^5 \text{ BTU/h}$$

$$T_{h,o} = T_{h,i} - \frac{Q}{\dot{m}c_{p,h}}$$

$$= 260 - \frac{600000}{100000} = 200°f$$

$$\Delta T_1 = T_{h,i} - T_{c,o} = 260 - 210 = 50°F$$
$$\Delta T_2 = T_{h,o} - T_{c,i} = 200 - 90 = 110°F$$

$$\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2/\Delta T_1)} = \frac{110 - 50}{\ln(110/50)} = 76.1°f$$

$$A = \frac{Q}{U_o \Delta T_{lm}} = \frac{600000}{150 \times 76.1} \approx 52.5 \text{ ft}^2$$

Out side area per tube of length L : $A_t = \pi D_o L$

30 tubes: $L = \dfrac{A}{N_t \pi D_o}$

$\Rightarrow \dfrac{52.5}{30 \times \pi \times 0.041667} = 13.4$ ft per tube

(b) For shorter tubes:

- Add tubes (increase N_t) or increase tube diameter to increase area per unit length
- Raise U_o → higher velocities, better shell baffles, roughened tubes, cleaner surfaces / lower fouling, different materials, or using enhanced / finned tubing
- Change temperature driving force → higher hot inlet temp or relax the cold-side outlet temp

2) Let's treat each sphere as lumped (uniform $T_s(t)$), with Al $k = 205$ W/m·K

$Bi = hL_c/k$, $L_c = R/3$

$Bi = \dfrac{500 \cdot (0.01/3)}{205} \approx 0.008 << 0.1$ → lumped is valid

$A_{total} = N(4\pi R^2) = \dfrac{m_s}{\rho}\dfrac{3}{R} = \dfrac{3m_s}{\rho R}$

$m_s = 1$ kg, $\rho = 2700$ kg/m^3, $R = 0.01$ m

$A = \dfrac{3(1)}{2700 \cdot 0.01} = 0.111$ m^2

Total heat capacity of shres: $C_S = m_s C_P$

$\Rightarrow C_S = 1 \cdot 900 = 900$ J/K

$hA = 500 \times 0.111 = 55.56$ W/K

(a) Water temperature fixed at $20^\circ C$
Newton cooling (lumped):

$$\frac{d\theta}{dt} = -\frac{hA}{C_S}\theta, \quad \theta = T_s - T_w, \quad T_w = 20^\circ C$$

$$\theta t = \theta_i e^{-t/\tau}, \quad \tau = \frac{C_S}{hA} = \frac{900}{55.56} = 16.2 \text{ s}$$

From $T_{s,i} = 90^\circ C$ to $T_{s,f} = 40^\circ C$

$$t = \tau \ln\left(\frac{T_{s,i} - 20}{T_{s,f} - 20}\right) = 16.2 \ln\left(\frac{90-20}{40-20}\right)$$

$t = \underline{20.3s}$

(b) Water temperature changes (perfecty mixed, $1L = 1kg$)
het $C_w = m_w C_{p,w} \simeq 1 \cdot 4180 = 4180$ J/K

Energy balances

$$C_S \frac{dT_s}{dt} = -hA(T_s - T_w)$$

$$C_w \frac{dT_w}{dt} = +hA(T_s - T_w)$$

Define $\theta = T_s - T_w$, then

$$\frac{d\theta}{dt} = -k\theta \Rightarrow k = hA\left(\frac{1}{C_s} + \frac{1}{C_w}\right)$$

$C_sT_s + C_wT_w = \text{const} \rightarrow$

$$T_s(t) = \frac{C_s T_{s,i} + C_w T_{w,i}}{C_s + C_w} + \frac{C_w}{C_s + C_w}\theta_0 e^{-kt}$$

with $\theta_0 = T_{si} - T_{wi} = 70K$

$$k = 55.56\left(\frac{1}{900} + \frac{1}{4180}\right) = 0.0751\ s^{-1}$$

$$\frac{C_w}{C_s + C_w} = \frac{4180}{5080} = 0.8236$$

$$T_{eq} = \frac{900(90) + 4180(20)}{5080} = 32.40°C$$

$$T_s(t) = 32.40 + 0.8236 \times 70e^{-0.0751t}$$

Setting $T_s(t) = 40°C$

$$\Rightarrow 40 - 32.40 = 57.65\,e^{-0.0751t}$$

$$\Rightarrow e^{-0.0751t} = 0.1318$$

$$\Rightarrow t = 26.7\,s$$